

EXP.NO-8	MINI PROJECT-Global Travel and Tourism Intelligence

AIM:

To clean, analyze, and visualize global tourism data using Python, Power BI, and Tableau to identify tourism trends, travel costs, destination patterns, ratings, and festival-related tourism insights.

DOMAIN:

Tourism Analytics / Travel Analytics

PROBLEM STATEMENT:

Raw tourism data contains multiple factors such as tourist arrivals, travel costs, ratings, seasons, festivals, and destination types. This project transforms the raw data into meaningful visual insights for easier tourism analysis and comparison.

Dataset Description:

The dataset contains 112 raw records covering 2022–2025 and 25 destinations across different countries and continents.

It includes information about:

1. Tourist arrivals
2. Travel and accommodation costs
3. Food and flight costs
4. Tourist ratings
5. Destination type
6. Peak season
7. Major festivals
8. Temperature
9. Visa difficulty

After cleaning, the final dataset contained 103 records and 18 columns.

PROCEDURE:

1. Loaded the raw CSV dataset using Python and Pandas.
2. Inspected the dataset structure, data types, missing values, and duplicates.
3. Removed duplicate records.
4. Filled missing travel-cost and tourist-rating values using median imputation.
5. Created derived fields such as:
6. Total Travel Cost
7. Cost Category
8. Tourism Growth %
9. Performed exploratory data analysis using Pandas.
10. Created static and interactive visualizations using Python.
11. Exported the cleaned dataset.
12. Created an interactive Global Tourism Intelligence Dashboard in Power BI.
13. Created an interactive Travel Explorer – Global Destination Analytics dashboard in Tableau.

Key Insights:

Asia recorded the highest tourist arrivals among the continents.

Tourist arrivals increased consistently from 2022 to 2025.

City & Culture destinations accounted for large portions of total tourist arrivals.

Luxury destinations had the highest average travel cost.

The relationship between travel cost and tourist rating was weakly positive.

Tourist arrivals varied significantly across destinations, with major global cities contributing substantial volumes.

Tourism patterns also varied across peak seasons, destination types, and festivals.

Python Analysis

Python was used for data cleaning, processing, exploratory data analysis, and visualization.

Data Cleaning

Loaded the dataset using Pandas.

Checked dataset structure, data types, missing values, and duplicate records.

Removed 9 duplicate records.

Filled missing Tourist_Rating and Avg_Travel_Cost_USD values using the median.

Final cleaned dataset: 103 rows × 18 columns.

Data Processing

The following derived columns were created:

Total_Travel_Cost_USD = Accommodation + Food + Flight costs

Cost_Category = Budget, Moderate, or Premium

Tourism_Growth_% = year-over-year growth for each destination

Exploratory Data Analysis

Python was used to analyze:

Tourist arrivals by country and destination

Yearly tourism trends

Tourist arrivals by continent

Average travel cost by destination type

Average tourist rating by destination

Festival-wise tourist arrivals

Peak-season tourism patterns

Visa difficulty and tourist arrivals

Relationship between travel cost and tourist rating

Relationship between temperature and tourist arrivals

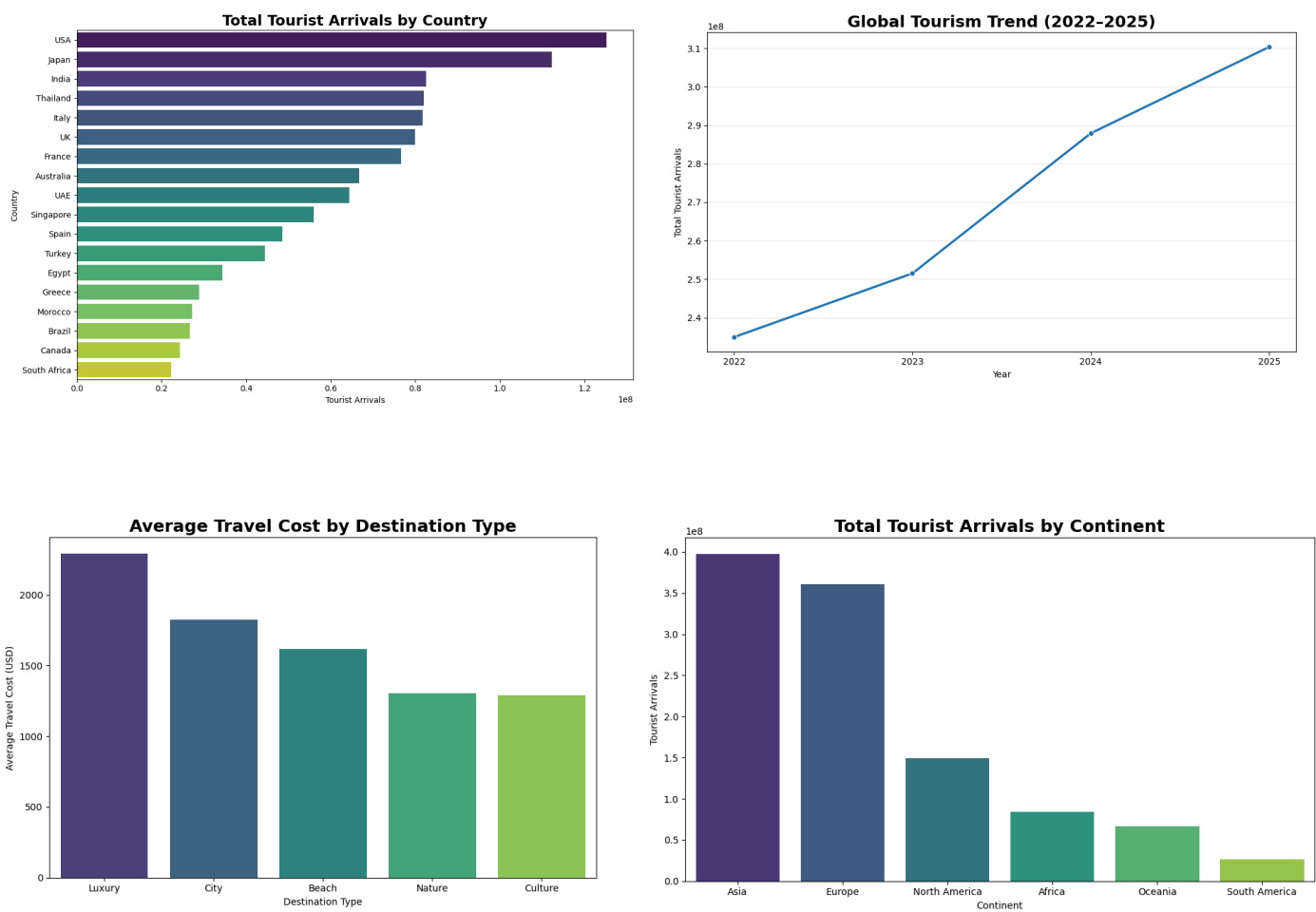
Python Visualizations

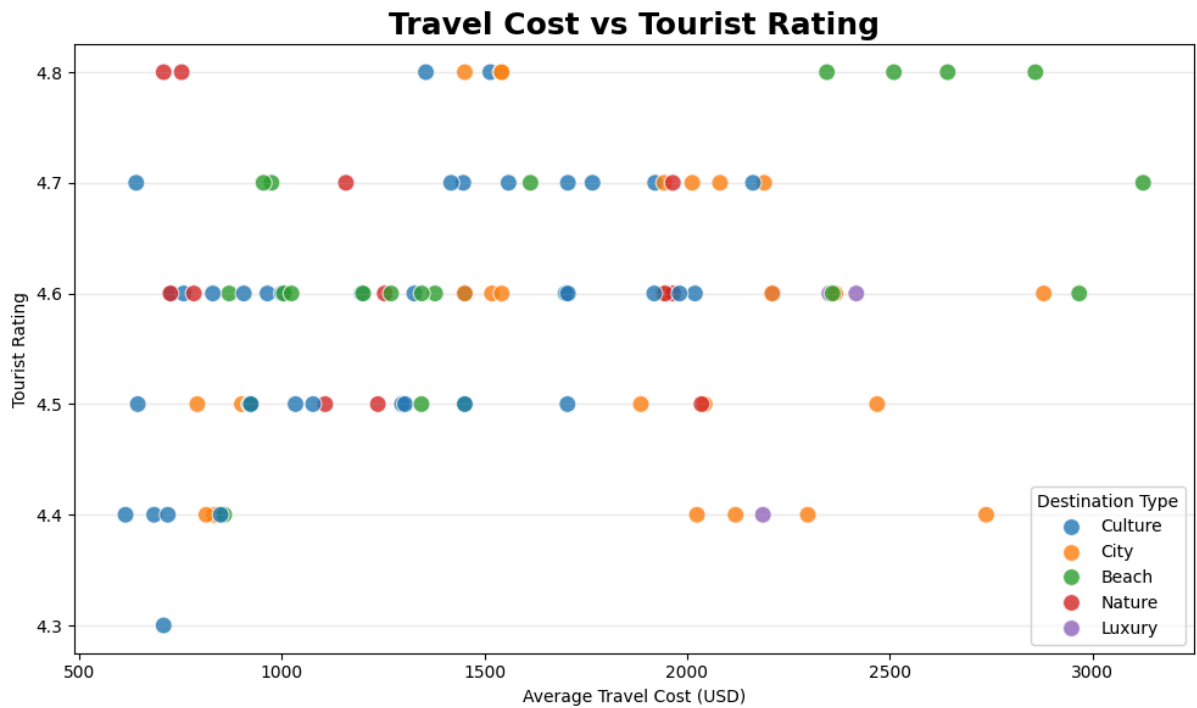
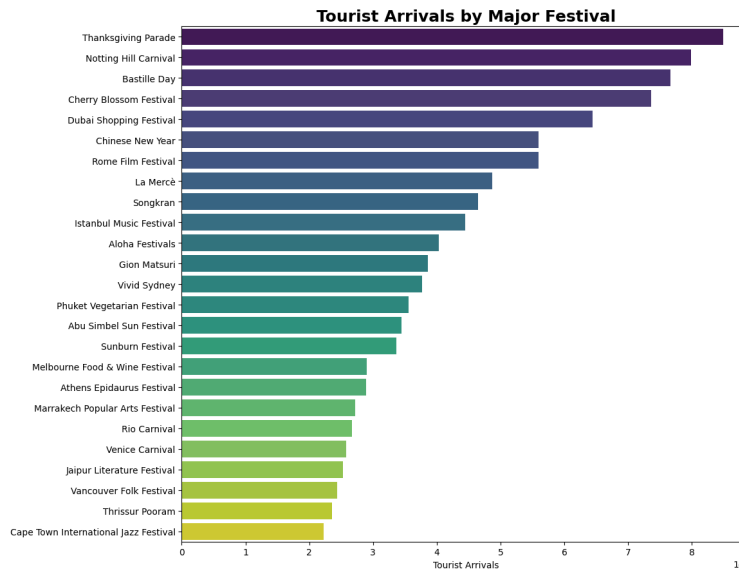
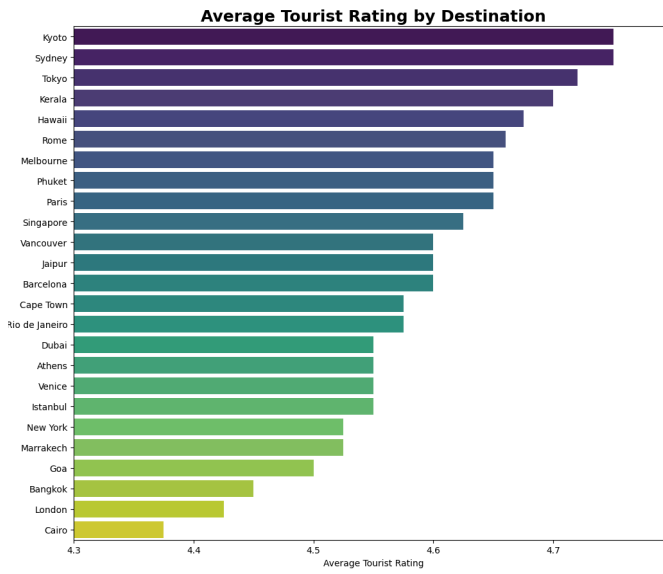
Visualizations were created using Matplotlib and Seaborn to identify major tourism patterns.

An interactive Plotly scatter plot was also created to explore the relationship between travel cost, tourist rating, destination type, and tourist arrivals.

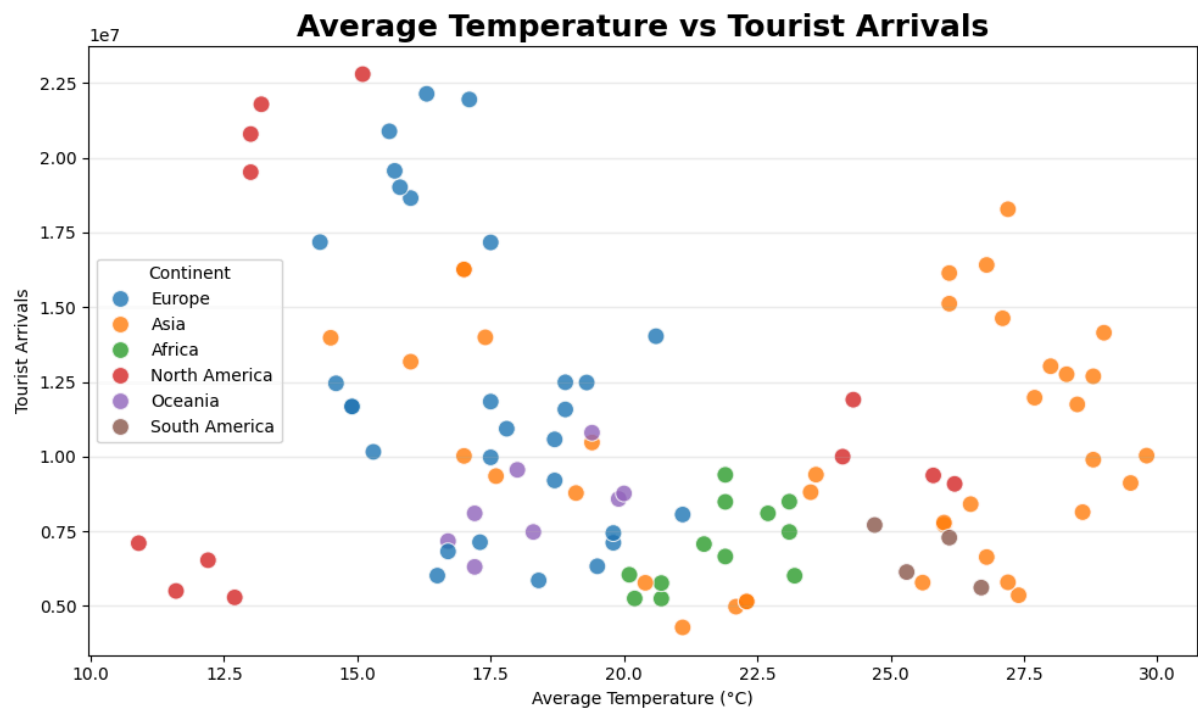
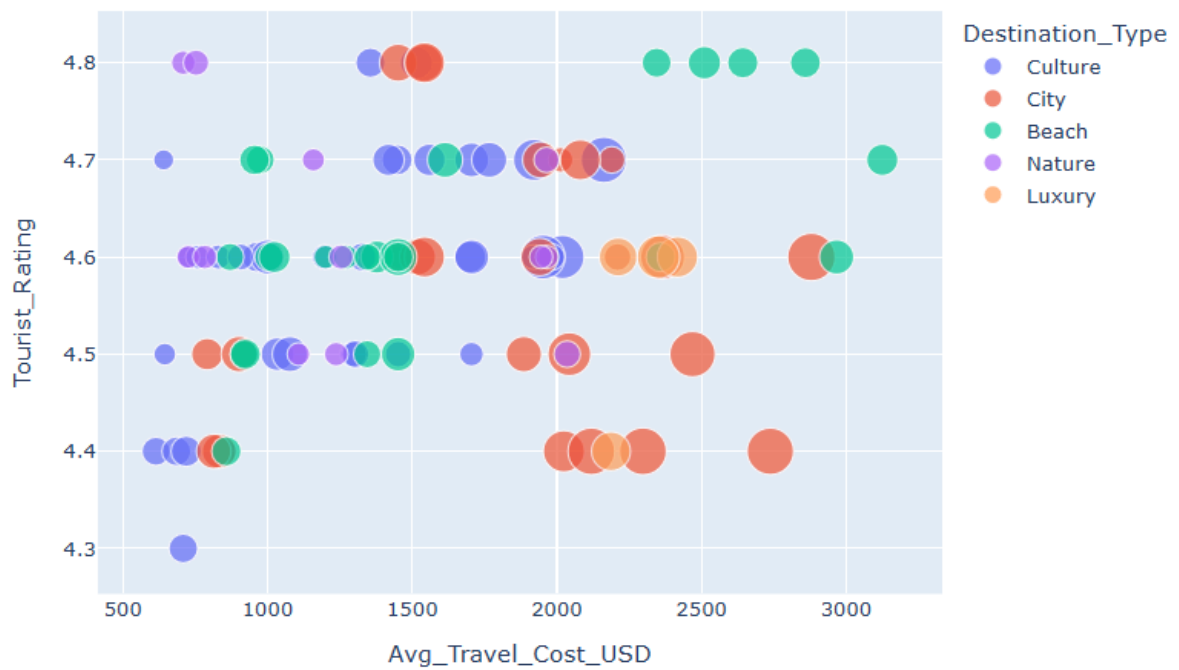
The processed dataset was finally exported as:

global_tourism_analytics_clean.csv





Interactive Travel Cost vs Tourist Rating



Power BI Output

The cleaned dataset was imported into Power BI to create an interactive Global Tourism Intelligence Dashboard. KPI cards and visualizations were used to analyze tourist arrivals, travel costs, ratings, yearly trends, destination types, continents, and the relationship between travel cost and tourist rating.

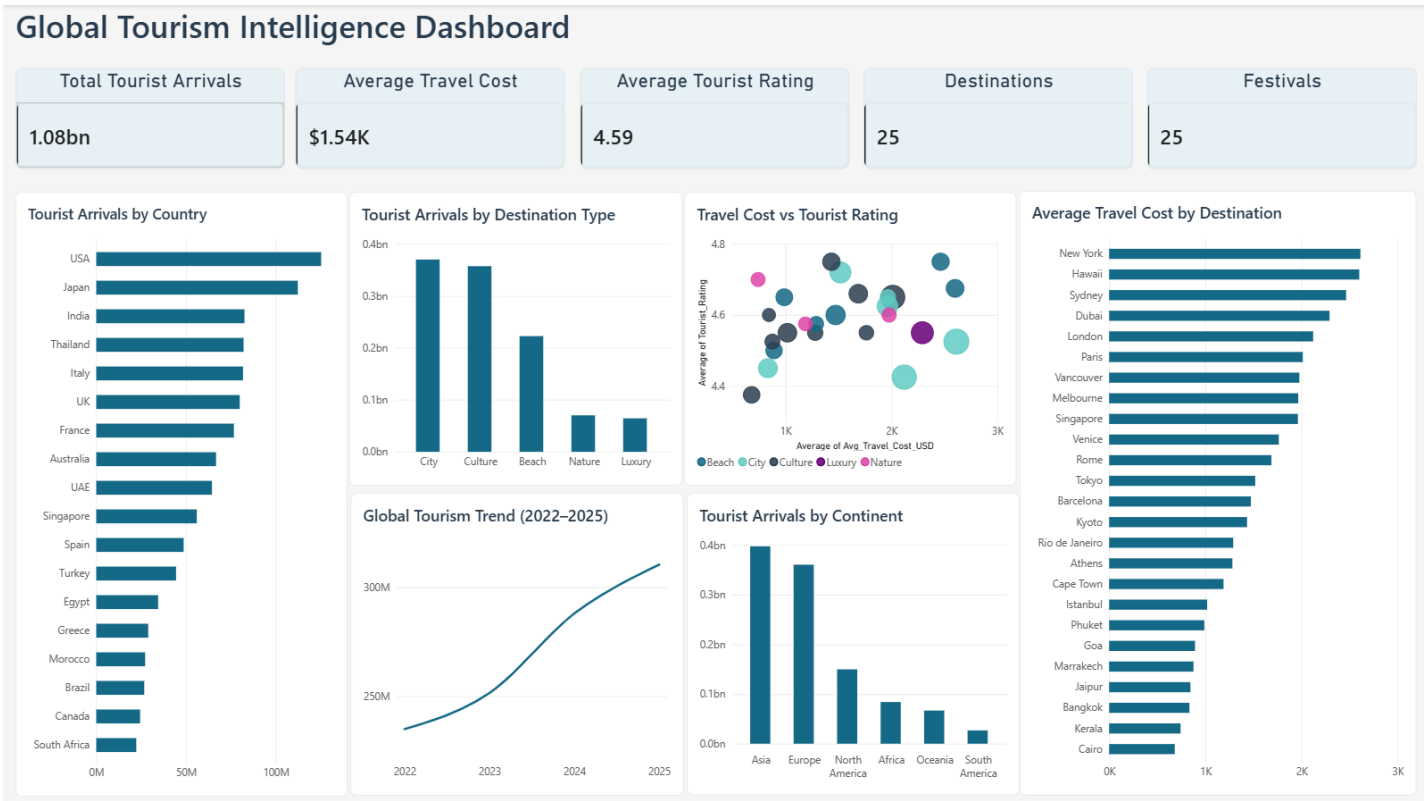
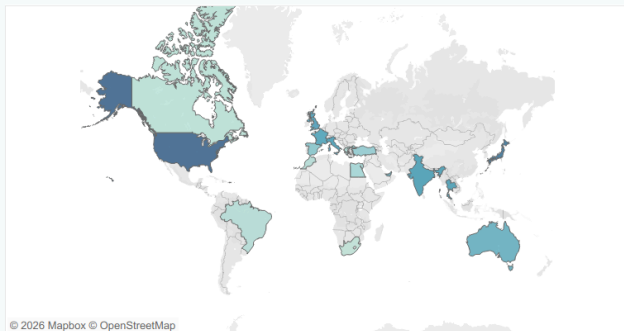


Tableau Output

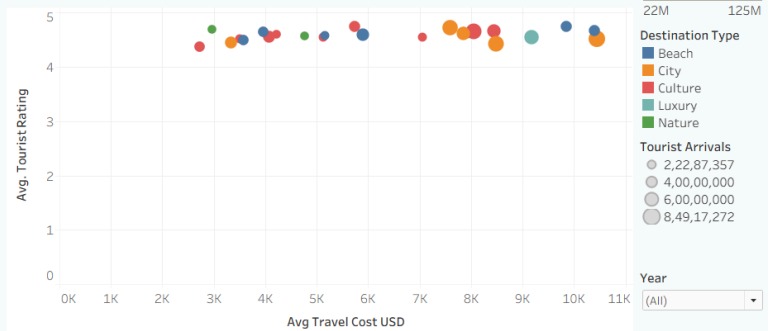
The cleaned dataset was also used in Tableau to create an interactive Travel Explorer – Global Destination Analytics dashboard. A geographic map, heatmap, scatter plot, and festival analysis were used along with Year, Destination, and Continent filters for interactive tourism exploration.

Travel Explorer — Global Destination Analytics

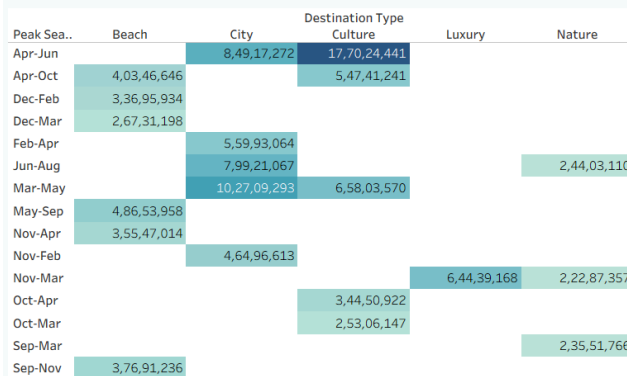
Global Tourism Map



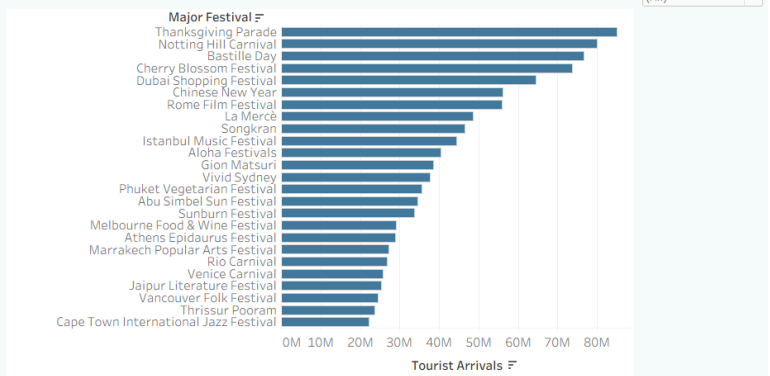
Destination Value Explorer



Season & Destination Type Heatmap



Festival Tourism Explorer



RESULT:

The project successfully transformed raw tourism data into a cleaned analytical dataset and interactive dashboards using **Python**, **Power BI**, and **Tableau**. The visualizations provide a clear understanding of global tourism trends, destination popularity, travel costs, ratings, seasons, and festival-related patterns.